MAXIMUM QUARTET DISTANCE

Abstract. Let M_n be the maximum quartet distance between two unrooted binary phylogenetic trees on a common n-element leaf set. We prove, for every $n \geq 4$,

$$\frac{2}{3}\binom{n}{4} \leq M_n \leq \min\left\{\binom{n}{4}, \frac{2}{3}\binom{n}{4} + \frac{n(6n^2-11n+6)}{9}\right\}.$$

Consequently, $M_n = (2/3 + O(n^{-1}))\binom{n}{4}$, proving the Bandelt–Dress conjecture. The proof combines a sharp inequality for eight dihedral four-point permutation patterns with random rotation systems of phylogenetic trees. Under the latter randomization, the quartet displayed by a tree is never the crossing split, while each of the other two splits is the crossing split with probability 1/2.

1. Introduction

An unrooted binary phylogenetic tree on a finite set X is a finite tree whose degree-one vertices are bijectively labelled by X and whose remaining vertices have degree three. For $Q \in \binom{X}{4}$, suppressing the degree-two vertices in the minimal subtree spanning Q gives one of the three quartet topologies on Q; denote it by $q_T(Q)$. The quartet distance between two such trees is

$$d_{\mathrm{Q}}(T_1, T_2) = \left|\{Q \in \tbinom{X}{4} : q_{T_1}(Q) \neq q_{T_2}(Q)\}\right|.$$

Writing M_n for the maximum of this quantity over pairs of trees on a common n-element leaf set, Bandelt and Dress conjectured that

$$M_n = \left(\frac{2}{3} + o(1)\right)\binom{n}{4} \qquad \text{as } n \to \infty$$

[3].

The conjecture has a substantial history. Alon, Snir, and Yuster proved the general upper bound $(9/10 + o(1))\binom{n}{4}$ and obtained finite improvements over the standard random-labelling lower bound [2]. Alon, Naves, and Sudakov subsequently improved the general upper bound to $(0.69 + o(1))\binom{n}{4}$ and proved the conjecture for pairs of caterpillar trees [1]. Chor, Erdős, and Komornik gave a strongly explicit construction with distance $(2/3 + o(1))\binom{n}{4}$, always exceeding the basic 2/3 benchmark in their family [5]. Snir, Weissberg, and Yuster studied the related problem when part of the leaf labelling is fixed [7]. The general conjecture was still recorded as open in recent work on metrics for phylogenetic tree space [6].

Permutation patterns already enter the caterpillar case. The proof of Alon, Naves, and Sudakov reduces that case to a 1/3 lower bound for the total density of the eight patterns

$$\{1234, 1243, 2134, 2143, 3412, 4312, 3421, 4321\}.$$

The eight patterns used in the present paper are different: they form the dihedral orbit of 1234. The new step is a randomized circular-order representation that applies to every binary phylogenetic tree, not only to caterpillars. Equal crossing splits of two circular orders are exactly detected by the dihedral pattern set, and random rotation systems connect those crossing splits to the quartets displayed by the original trees.

Theorem 1.1. *For every integer $n \geq 4$,*

$$\frac{2}{3}\binom{n}{4} \leq M_n \leq \min\left\{\binom{n}{4}, \frac{2}{3}\binom{n}{4} + \frac{n(6n^2-11n+6)}{9}\right\}.$$

Equivalently, the nontrivial upper estimate may be written as

$$\frac{M_n}{\binom{n}{4}} \leq \frac{2}{3} + \frac{8(6n^2-11n+6)}{3(n-1)(n-2)(n-3)} = \frac{2}{3} + \frac{16}{n} + \frac{200}{3n^2} + O(n^{-3}).$$

In particular,

$$M_n = \left(\frac{2}{3} + O\left(\frac{1}{n}\right)\right)\binom{n}{4}.$$

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Here is the proof in outline. A circular order on four leaves selects the split joining opposite positions, which we call its crossing split. A theorem of Chan, Kráľ', Noel, Pehova, Sharifzadeh, and Volec implies that any two circular orders on n elements have the same crossing split on at least $(1/3 - O(1/n))\binom{n}{4}$ four-subsets. On the other hand, choose a uniform random cyclic order of the three incident edges at every internal vertex of a phylogenetic tree. For any fixed quartet, the split displayed by the tree is then the unique split that can never be the crossing split; the other two splits occur with probability $1/2$ each. Thus, if two trees disagree on a proportion d of their quartets, their random circular orders have the same crossing split on an expected proportion

$$\frac{1}{2}(1-d) + \frac{1}{4}d = \frac{1}{2} - \frac{d}{4}.$$

Comparison with the circular-order lower bound gives $d \le 2/3 + O(1/n)$.

The sole non-elementary input is the published permuton inequality from [4, Theorem 7]. We give the complete reduction from that inequality to Theorem 1.1, including the finite-size error. The argument never assumes independence among the quartet variables associated with different four-subsets; it uses only the one-quartet distribution and linearity of expectation.

2. Circular orders and permutation patterns

By a circular order on X we mean an equivalence class of linear arrangements under cyclic shifts and reversal. If the restriction of a circular order C to $Q = \{a, b, c, d\}$ has representative (a, b, c, d), define its crossing split by $\chi_C(Q) = ac \mid bd$. This is well defined because cyclic shifts and reversal preserve the pair of opposite positions. For two circular orders C, C' on X, put

$$K(C, C') = \left|\{Q \in \tbinom{X}{4} : \chi_C(Q) = \chi_{C'}(Q)\}\right|.$$

Let

$$\mathcal{D} = \{1234, 1432, 2143, 2341, 3214, 3412, 4123, 4321\} \subset S_4.$$

Thus $\mathcal{D}$ consists of the four cyclic shifts of 1234 and their reversals. Equivalently, it is the dihedral subgroup of S_4 generated by 2341 and 4321; in particular, $\mathcal{D}$ is inverse-closed.

Lemma 2.1. *Let C, C' be circular orders on an n-element set. Choose representatives $C = (x_1, \dots, x_n)$ and C', and let $\pi(i)$ be the position of x_i in the chosen representative of C'. If $i_1 < i_2 < i_3 < i_4$ and $Q = \{x_{i_1}, x_{i_2}, x_{i_3}, x_{i_4}\}$, then*

$$\chi_C(Q) = \chi_{C'}(Q) \iff \operatorname{pat}\big(\pi(i_1), \pi(i_2), \pi(i_3), \pi(i_4)\big) \in \mathcal{D}.$$

Proof. On four labelled elements there are exactly three circular orders and exactly three splits into two unordered pairs. Sending a circular order to the split formed by opposite positions is a bijection. Consequently, two restricted circular orders have the same crossing split precisely when they differ by a cyclic shift or reversal.

Set

$$\sigma = \operatorname{pat}\big(\pi(i_1), \pi(i_2), \pi(i_3), \pi(i_4)\big).$$

The order in which the four elements occur in C' is described by σ^{-1}. Hence the two restricted circular orders agree up to a cyclic shift or reversal precisely when $\sigma^{-1} \in \mathcal{D}$. Since $\mathcal{D}$ is inverse-closed, this is equivalent to $\sigma \in \mathcal{D}$. ☐

A permuton is a Borel probability measure μ on $[0, 1]^2$ with uniform marginals. For $\sigma \in S_k$, its density $d(\sigma, \mu)$ is the probability that, after k independent points sampled from μ are ordered by increasing first coordinate, the relative order of their second coordinates is σ. Uniform marginals imply that ties occur with probability zero.

Theorem 2.2 (Chan–Kráľ'–Noel–Pehova–Sharifzadeh–Volec)**.** *For every permuton μ,*

$$\sum_{\sigma \in \mathcal{D}} d(\sigma, \mu) \ge \frac{1}{3}.$$

This is [4, Theorem 7]; its equality characterization is not needed here.

Write

$$(n)_4 = n(n-1)(n-2)(n-3), \qquad \alpha_n = \frac{(n)_4}{n^4} = \frac{(n-1)(n-2)(n-3)}{n^3}.$$

Lemma 2.3. *For any two circular orders C, C' on an n-element set, where $n \ge 4$,*

$$K(C, C') \ge \left(1 - \frac{2}{3\alpha_n}\right)\binom{n}{4}.$$

Proof. Let $\pi \in S_n$ be the relative permutation from Lemma 2.1, and set

$$I_i = \left[\frac{i-1}{n}, \frac{i}{n}\right) \quad (1 \le i \le n),$$

with the harmless convention that $1 \in I_n$. Define the step permuton μ_π to have density n on

$$\bigcup_{i=1}^{n} I_i \times I_{\pi(i)}$$

and density zero elsewhere. For almost every first coordinate, the density integrates to one along the corresponding vertical section. The analogous statement holds horizontally, so both marginals are uniform.

Sample four independent points from μ_π, let Σ be their induced four-point pattern, and let E be the event that their first coordinates lie in four distinct intervals I_i. The four strip indices are independent and uniform on $[n]$, and therefore $\mathbb{P}(E) = \alpha_n$. Conditional on E, their increasing sequence is the increasing enumeration of a uniformly random four-subset of $[n]$. Since π is injective, distinct first-coordinate strips correspond to distinct second-coordinate strips; consequently, the within-strip coordinates cannot change the induced pattern. Thus Σ is the four-point pattern of π on that uniformly random four-subset. Lemma 2.1 gives

$$p := \mathbb{P}(\Sigma \in \mathcal{D} \mid E) = \frac{K(C, C')}{\binom{n}{4}}.$$

If $r := \mathbb{P}(\Sigma \in \mathcal{D} \mid E^c)$, then $0 \le r \le 1$. Theorem 2.2 consequently implies

$$\frac{1}{3} \le \alpha_n p + (1-\alpha_n) r \le \alpha_n p + 1 - \alpha_n, \qquad p \ge 1 - \frac{2}{3\alpha_n}.$$

Multiplying by $\binom{n}{4}$ proves the assertion. □

3. Random rotation systems

At every internal vertex of an unrooted binary phylogenetic tree T, choose one of the two cyclic orders of its three incident edges. Such a collection ρ is a rotation system, or equivalently an oriented ribbon structure, on T. A regular neighbourhood of the resulting ribbon tree is an oriented surface that deformation retracts onto T. Since T is a tree, this neighbourhood is a disk. The leaf caps therefore occur along its boundary in a cyclic order; after forgetting the orientation, write this circular order as C_ρ. Choosing the two cyclic orders independently and uniformly at all internal vertices gives the random circular order C_T.

Lemma 3.1. *For every deterministic rotation system ρ on T and every edge e, the leaves in each component of $T - e$ form a circular interval in C_ρ.*

Proof. Cut the regular neighbourhood of T across the rectangular band corresponding to e. The thickening of either component of $T - e$ meets the original boundary in one arc whose endpoints lie on the two sides of that band. All leaf caps belonging to that component occur on this arc, and no other leaf cap does. They are therefore consecutive in the induced circular order. □

For $Y \subseteq X$, let $T[Y]$ be the minimal connected subtree of T spanning Y, and let $T|Y$ be obtained from $T[Y]$ by suppressing every degree-two vertex. The ribbon structure on T induces one on $T|Y$: delete the unused incident bands, and replace each maximal chain of degree-two vertices by a single band. Notice that a cyclic order on two half-edges is unique, so no additional choice or reversal is introduced by this suppression.

Lemma 3.2 (Restriction of the boundary order)**.** *For every rotation system ρ on T and every subset Y of its leaves with $|Y| \ge 2$, the boundary order of the induced ribbon tree $T|Y$ is the restriction $C_\rho|Y$.*

Proof. Delete the components of $T - T[Y]$ one at a time. Each deleted component attaches to the surviving subtree through a single edge. By Lemma 3.1, its leaf caps occupy one boundary interval. Removing its thickening and closing the exposed boundary therefore deletes exactly that interval from the cyclic list of leaf caps and does not change the relative cyclic order of the surviving caps. After all such components have been removed, the boundary order is $C_\rho|Y$.

It remains to suppress degree-two vertices. Locally, the thickening of a degree-two vertex consists of a vertex disk joined to two bands. Replacing this disk and the two bands by one band is an orientation-preserving isotopy away from the leaf caps. It therefore does not change their cyclic order. Repeating this operation gives the asserted boundary order on $T|Y$. □

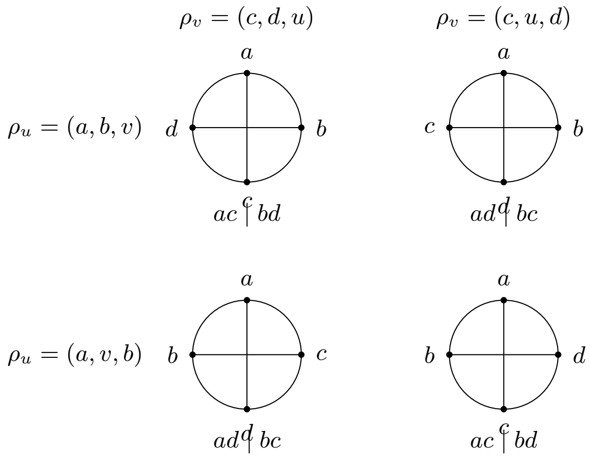

FIGURE 1. The four rotation choices for an induced quartet with true split $ab \mid cd$. In each circle the labels are shown in boundary order, and the two diameters join opposite positions; the split below the circle is therefore the crossing split.

Lemma 3.3. *Fix $Q \in \binom{X}{4}$ and a quartet topology s on Q. For the random circular order C_T,*

$$\mathbb{P}\big(\chi_{C_T}(Q) = s\big) = \begin{cases} 0, & s = q_T(Q), \\ \frac{1}{2}, & s \neq q_T(Q). \end{cases}$$

Proof. Write $q_T(Q) = ab \mid cd$. An edge on the path between the two branch vertices of the induced quartet separates $\{a, b\}$ from $\{c, d\}$. By Lemma 3.1, each pair is a circular interval in $C_T|Q$. Hence $ab \mid cd$ cannot be the crossing split.

By Lemma 3.2, the restricted circular order is the boundary order of the induced ribbon quartet $T|Q$. This quartet has two branch vertices u, v, where u is incident with the paths to a, b, v and v is incident with the paths to c, d, u. Using cyclic notation, the choices at u are (a, b, v) and (a, v, b), while those at v are (c, d, u) and (c, u, d). The four pairs of choices are equally likely: u and v are distinct internal vertices of the original tree, so their induced cyclic orders remain independent uniform choices. Rotations at vertices suppressed to degree two have no effect, since a cyclic order on two half-edges is unique. A boundary traversal gives the four outcomes shown in Figure 1.

Thus the two topologies distinct from $ab \mid cd$ each occur as the crossing split in exactly two of the four cases, and hence with probability 1/2. □

4. Proof of the theorem

Proof of Theorem 1.1. Let T_1, T_2 be binary phylogenetic trees on the same n-element leaf set, and put $N = \binom{n}{4}$. Let A be the number of four-subsets on which the trees agree and let

$$D = d_{\mathrm{Q}}(T_1, T_2) = N - A.$$

Choose the rotation systems of T_1 and T_2 independently, obtaining circular orders C_1, C_2.

Suppose first that $q_{T_1}(Q) = q_{T_2}(Q)$. By Lemma 3.3, the random variables $\chi_{C_1}(Q)$ and $\chi_{C_2}(Q)$ are independent uniform choices from the same two-element set, and hence agree with probability 1/2. If the two true quartet topologies differ, their two supports have exactly one topology in common, so the crossing splits agree with probability 1/4. Linearity of expectation gives

$$\mathbb{E}K(C_1, C_2) = \frac{1}{2}A + \frac{1}{4}D = \frac{N + A}{4}.$$

No independence between the contributions of distinct four-subsets is used here.

Lemma 2.3 holds for every realization of (C_1, C_2) and therefore also after taking expectations. Using $\alpha_n = (n-1)(n-2)(n-3)/n^3$, we obtain

$$\frac{N + A}{4} \geq \left(1 - \frac{2}{3\alpha_n}\right) N, \qquad D = N - A \leq \left(\frac{8}{3\alpha_n} - 2\right) N.$$

Since

$$1 - \alpha_n = \frac{6n^2 - 11n + 6}{n^3}, \qquad \frac{N}{\alpha_n} = \frac{n^4}{24},$$

the last bound becomes

$$D \le \frac{2}{3}N + \frac{8(1-\alpha_n)}{3\alpha_n}N = \frac{2}{3}\binom{n}{4} + \frac{n(6n^2-11n+6)}{9}.$$

Together with the trivial inequality $D \le N$, this proves the stated upper bound. Dividing the nontrivial estimate by N gives

$$\frac{D}{N} \le \frac{2}{3} + \frac{8(6n^2-11n+6)}{3(n-1)(n-2)(n-3)} = \frac{2}{3} + \frac{16}{n} + \frac{200}{3n^2} + O(n^{-3}).$$

For the lower bound, we include the standard random-labelling argument. Fix any two unlabelled binary tree shapes with n leaves, label the first by X, and label the second by a uniformly random bijection from X to its leaves. For a fixed $Q \in \binom{X}{4}$, condition on the four leaf positions receiving the labels in Q. Each of the three labelled quartet topologies is realized by exactly eight of the $4! = 24$ possible assignments of the four labels to those positions. The quartet induced by the randomly labelled tree therefore disagrees with the fixed quartet induced by the first tree with probability $2/3$. Consequently,

$$\mathbb{E}d_{\mathrm{Q}}(T_1, T_2) = \frac{2}{3}\binom{n}{4}.$$

Some labelling must have distance at least this expectation. Combining the lower and upper bounds proves the theorem. $\square$

Remark 4.1. The $O(n^3)$ additive term in the upper bound comes entirely from the event that two of the four samples in the step-permuton construction occupy the same first-coordinate strip. The estimate $\mathbb{P}(\Sigma \in \mathcal{D} \mid E^c) \le 1$ deliberately discards all structure on that collision event. A finer classification of the collision types could improve the finite error without changing the rotation-system part of the proof. Correlations among distinct four-subsets of leaves play no role.

Remark 4.2. The explicit estimate is intended to display the convergence rate, not to optimize moderate values of n. The minimum with the trivial bound is therefore included in Theorem 1.1. The nontrivial term first improves $M_n \le \binom{n}{4}$ at $n = 53$, and its normalized coefficient first drops below 0.69 at $n = 690$.

5. Formal verification

The reduction in this paper has been formalized in Lean 4 with mathlib; the source is available at `https://github.com/rvazdev-ex/quartet_distance_bandelt_dress`. The development independently defines the conventional finite graph model used here, proves it label-preservingly equivalent to the recursive tree representation used for counting, and, for $n \ge 4$, proves equality of the resulting quartet distances and maxima. It checks the finite bounds of Theorem 1.1, the normalized expansion, and the asymptotic conclusion. The inequality in Theorem 2.2 is supplied as an explicit hypothesis, not as a Lean axiom: the formalization verifies every step of the reduction from that cited result, but does not reprove the cited result itself. The audited endpoints contain no unproved placeholders or custom axioms and use only Lean's standard logical foundations.

6. Disclosure

Apart from minor manual edits, the proof and paper were completely generated by ChatGPT Sol 5.6 Pro. The proof was thoroughly checked by hand, step by step.

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